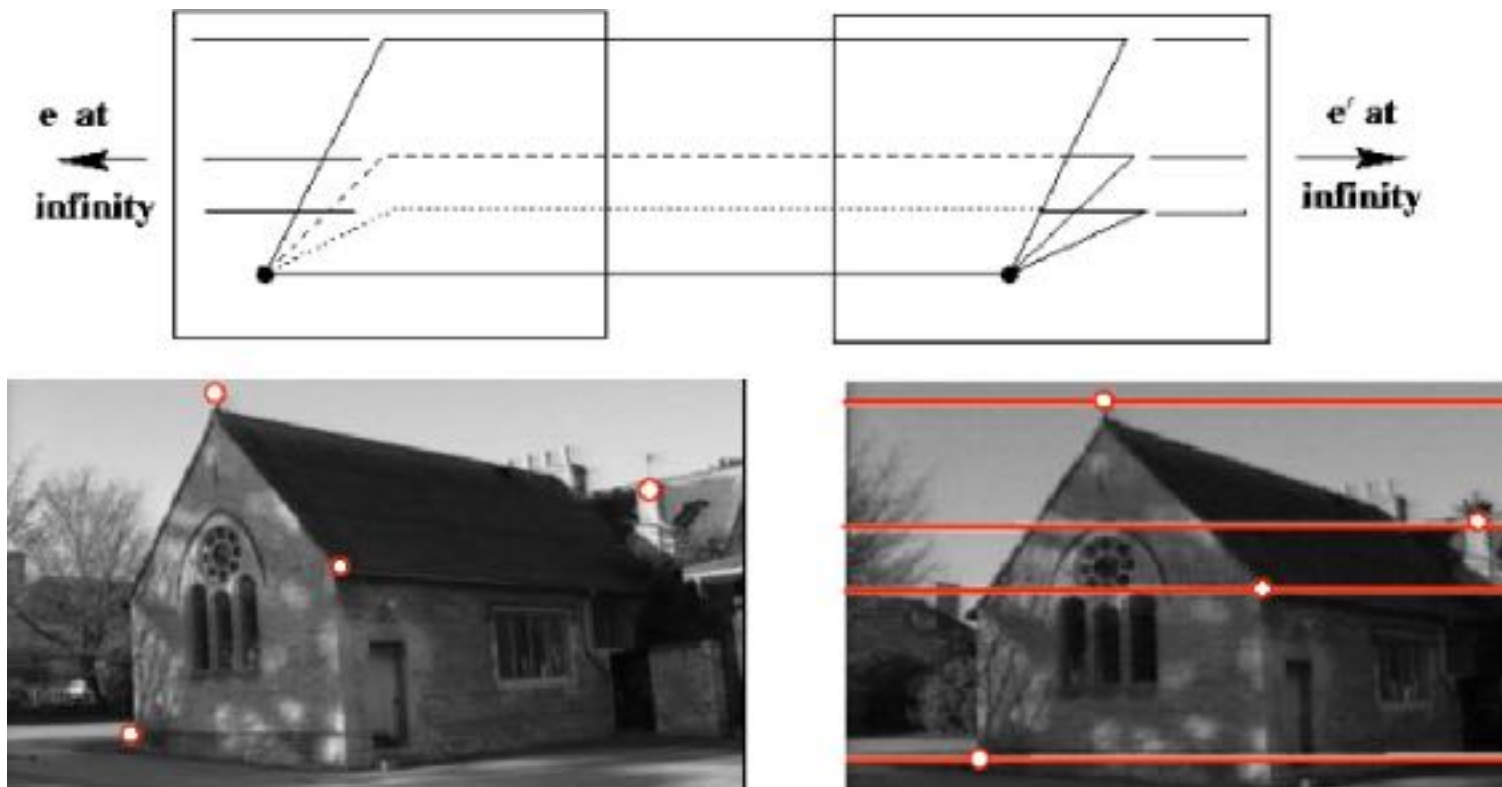


**Last week: stereo disparity
for inferring depth**

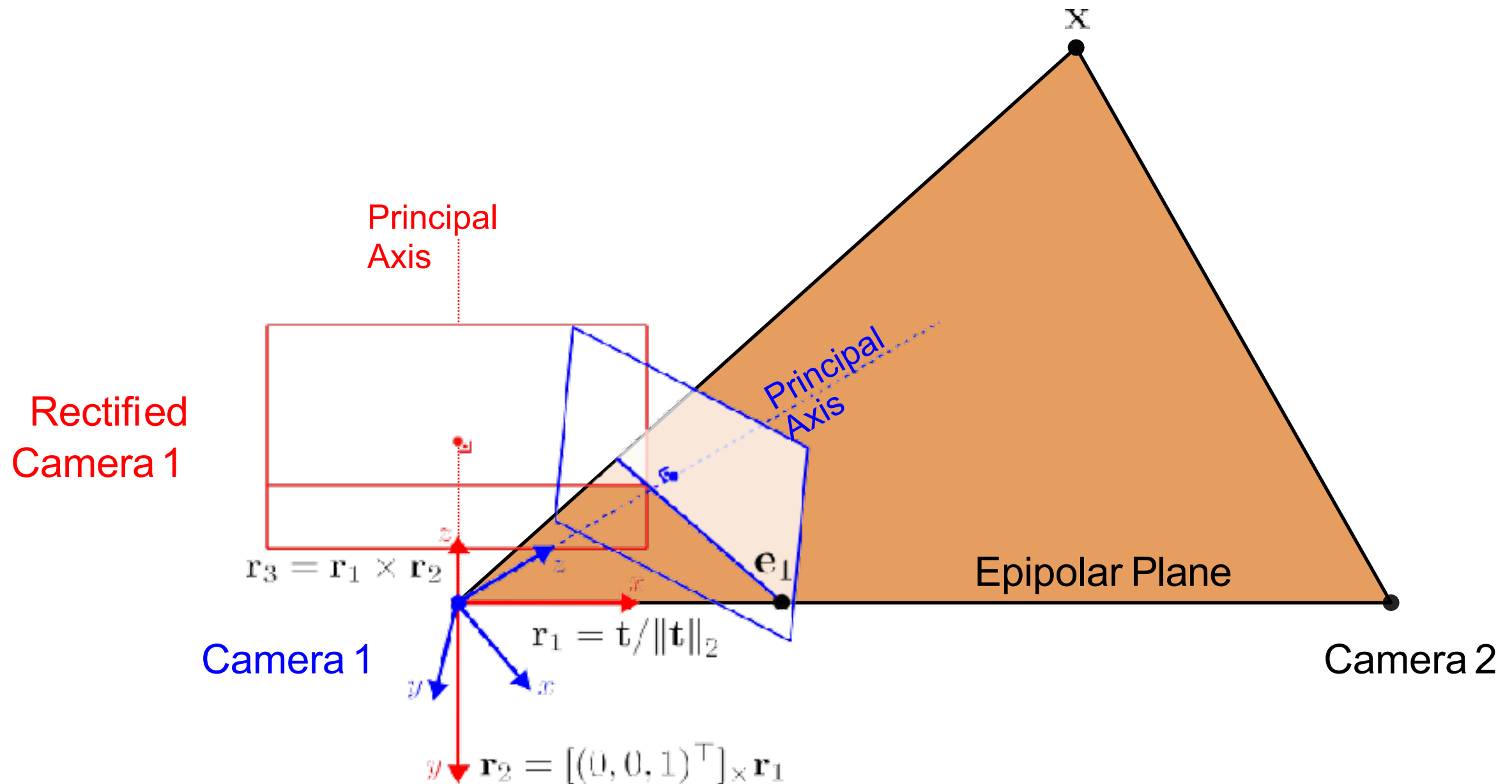
Rectification

What if both cameras face exactly the same direction?



- Image planes are co-planar $\Rightarrow$ Epipoles at infinity, epipolar lines parallel
- Correspondences search along horizontal scanlines (simplifies implementation)

We can enforce a rectified pair by rewarping one camera



Rectification: Example



- Correspondences are located on the **same image row** as the querypoint

Disparity to depth

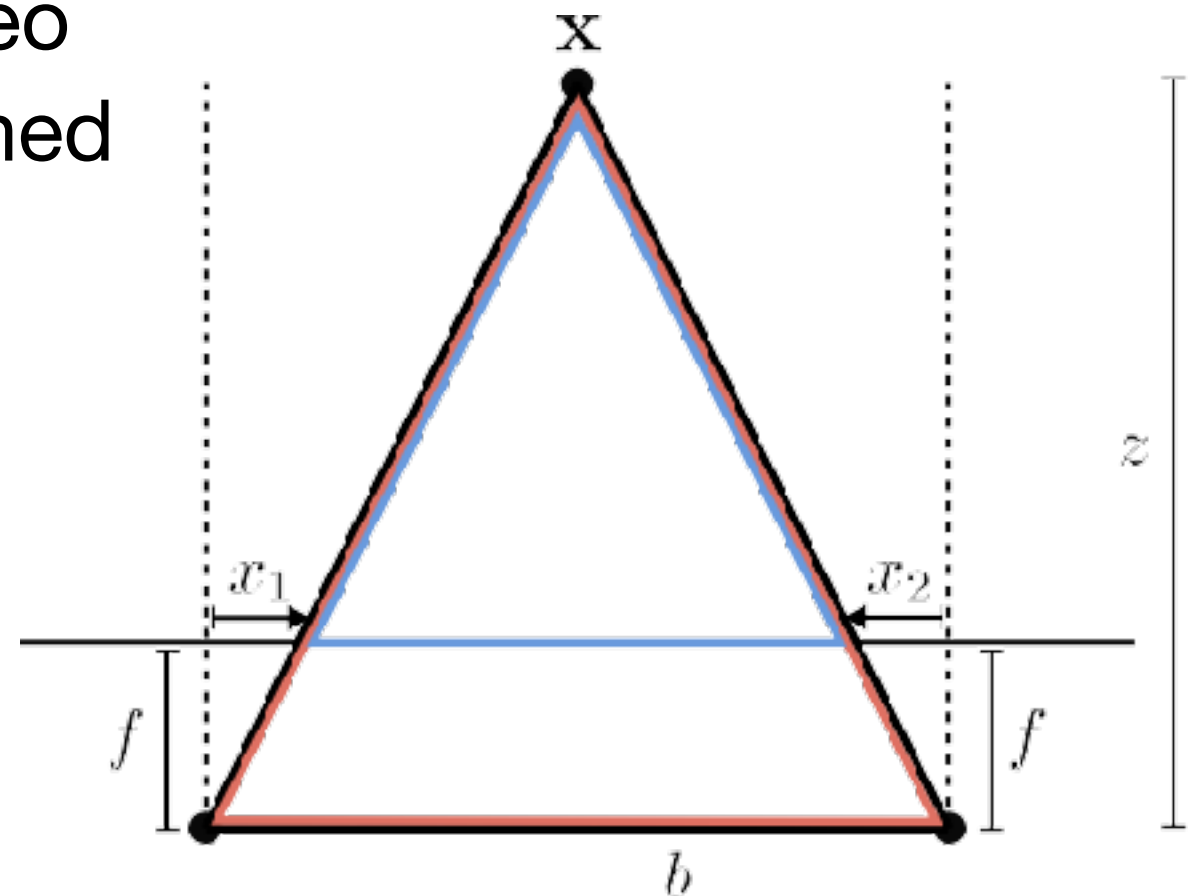
The left and right ray intersect
(shared epipolar plane).

Assume the disparity is $d = x_1 - x_2$.
Given a focal length f and the stereo
baseline b , the depth z is constrained
by

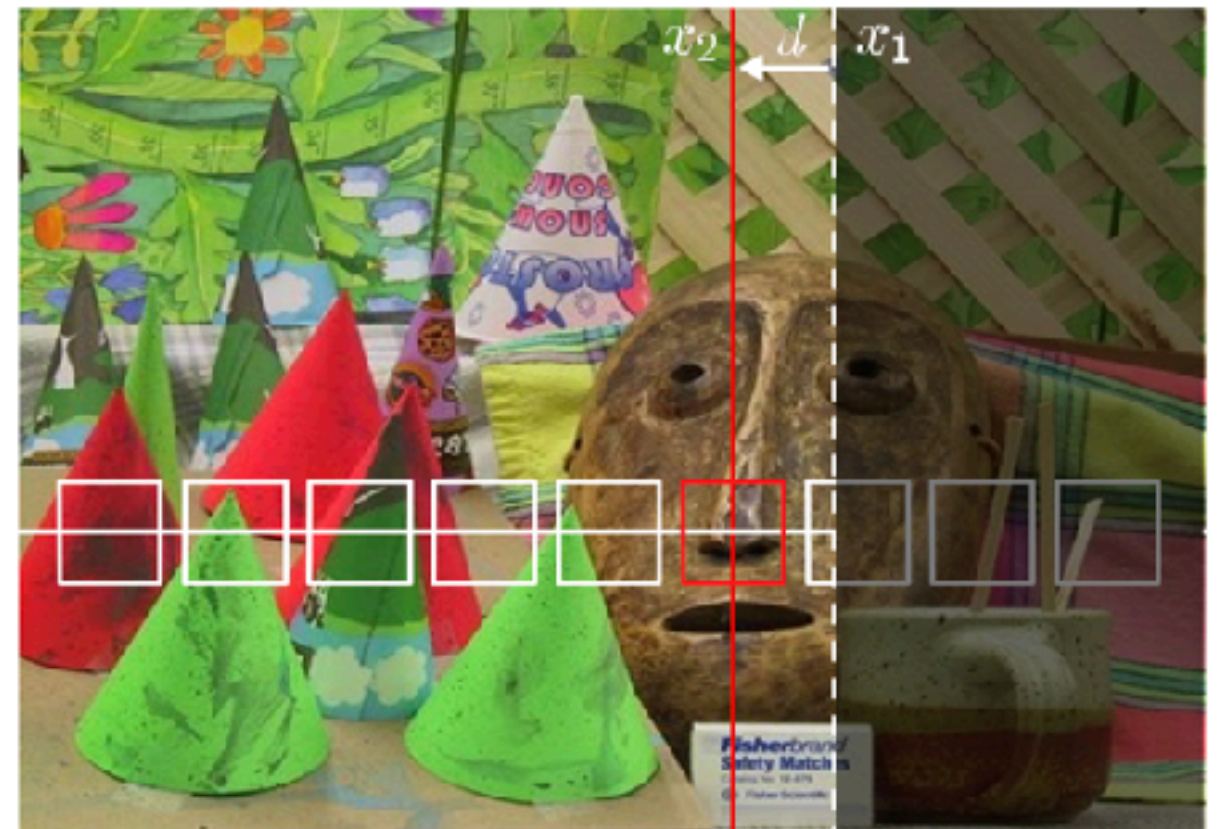
$$\frac{z - f}{b - d} = \frac{z}{b}$$

$$zb - fb = zb - zd$$

$$z = \frac{fb}{d}$$



Stereo matching

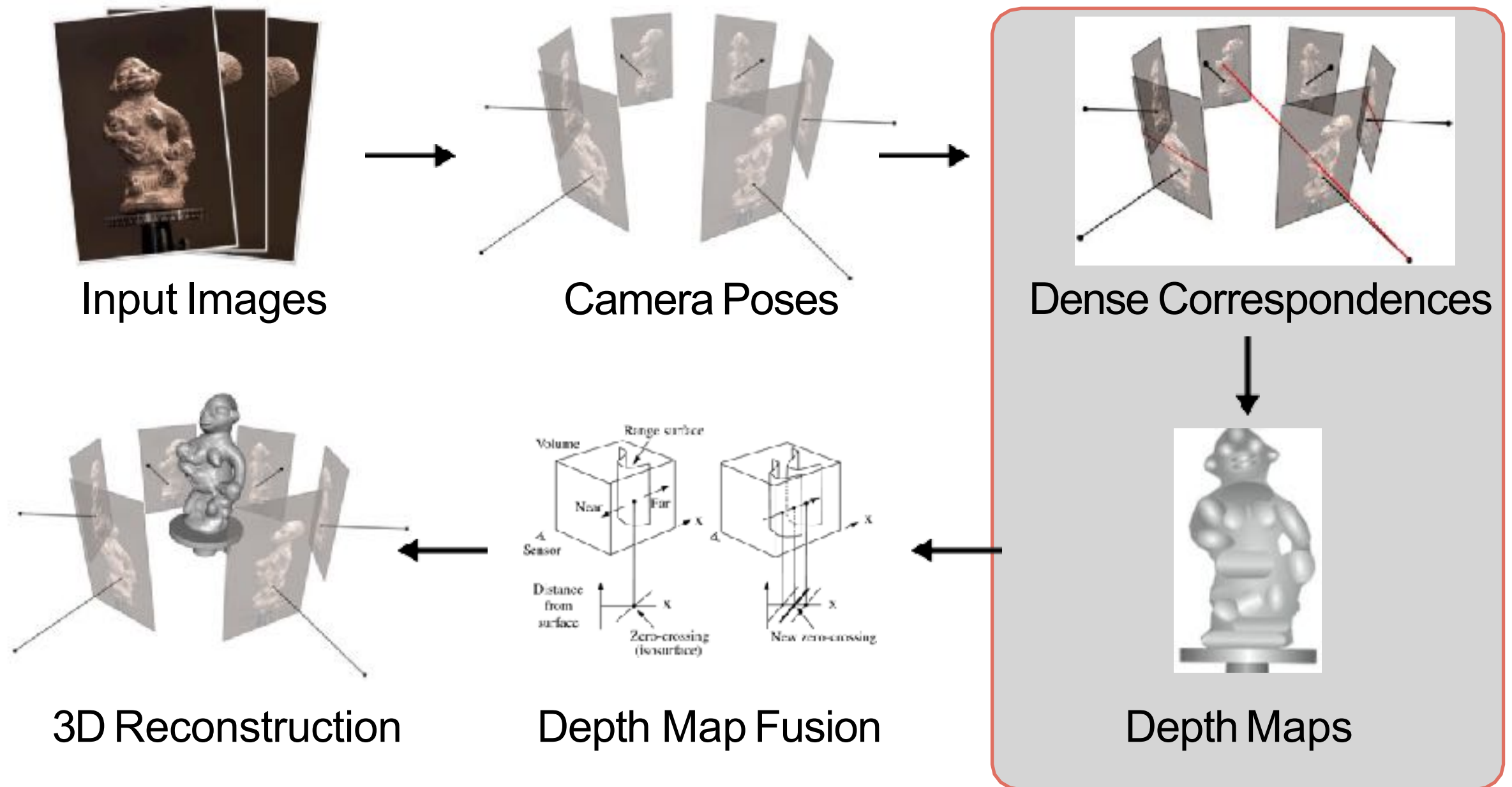


Block matching

- Choose disparity range
- For all pixels, compute best disparity
- Repeat in opposite direction, remove inconsistencies to reduce outliers



3D Reconstruction Pipeline





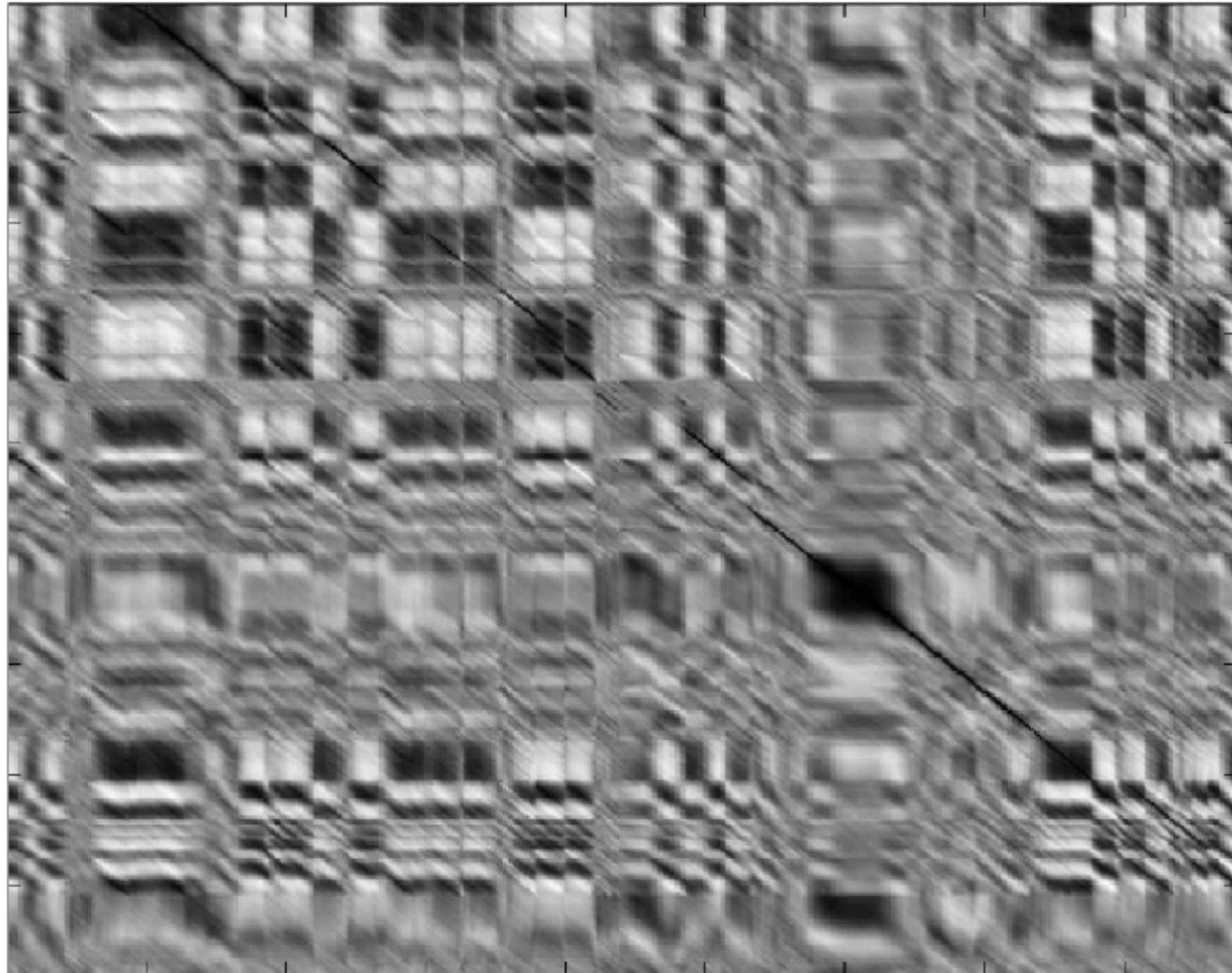
Despite all these problems: The „real“ depth is in practice mostly smooth - except at typically rare discontinuities such as object boundaries.

**Last week: Dynamic
programming for stereo
matching**

Disparity Space Image

Left scanline

Right scanline



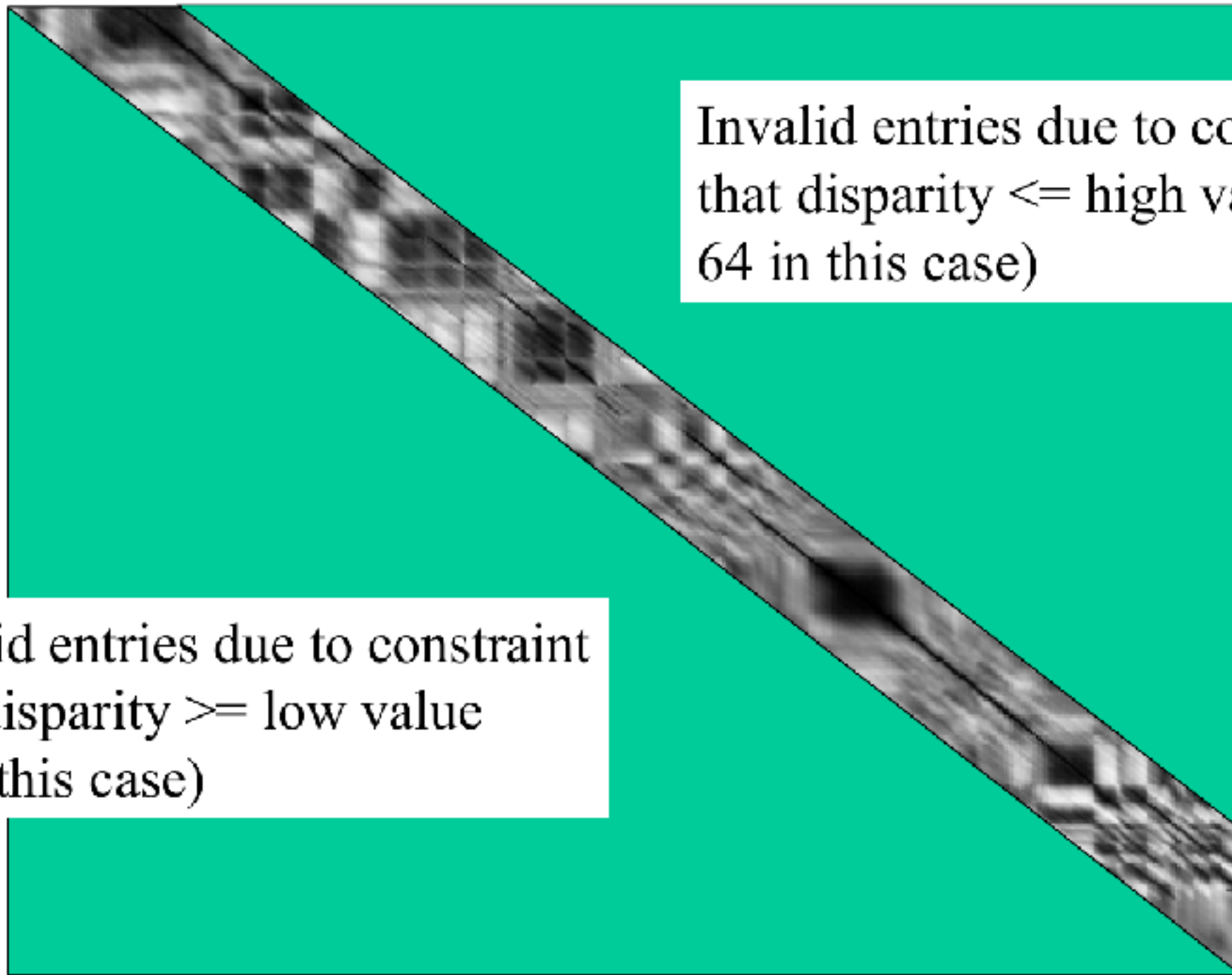
Disparity Space Image

Left scanline

Right scanline

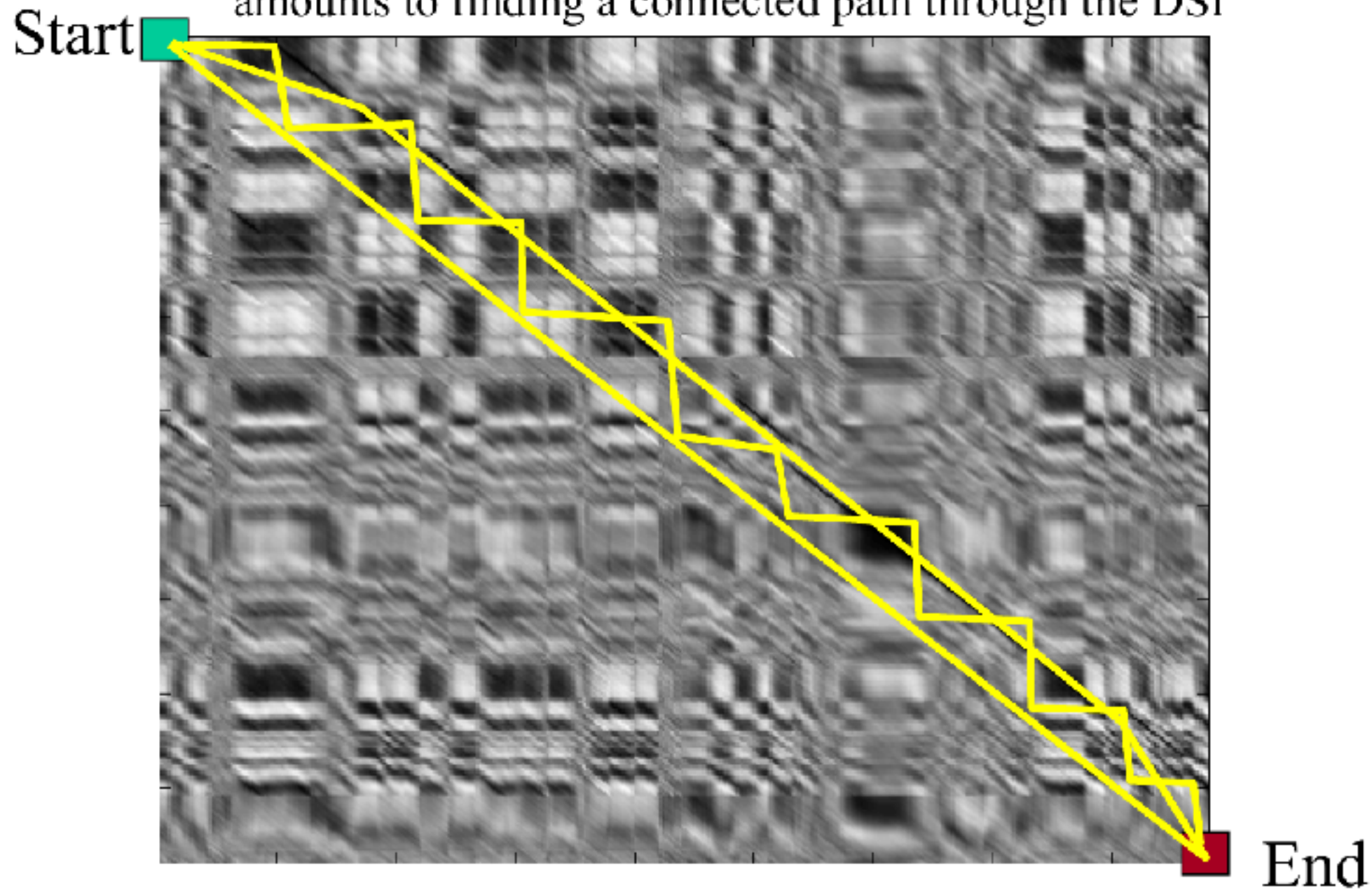
Invalid entries due to constraint
that disparity $\leq$ high value
64 in this case)

Invalid entries due to constraint
that disparity $\geq$ low value
(0 in this case)



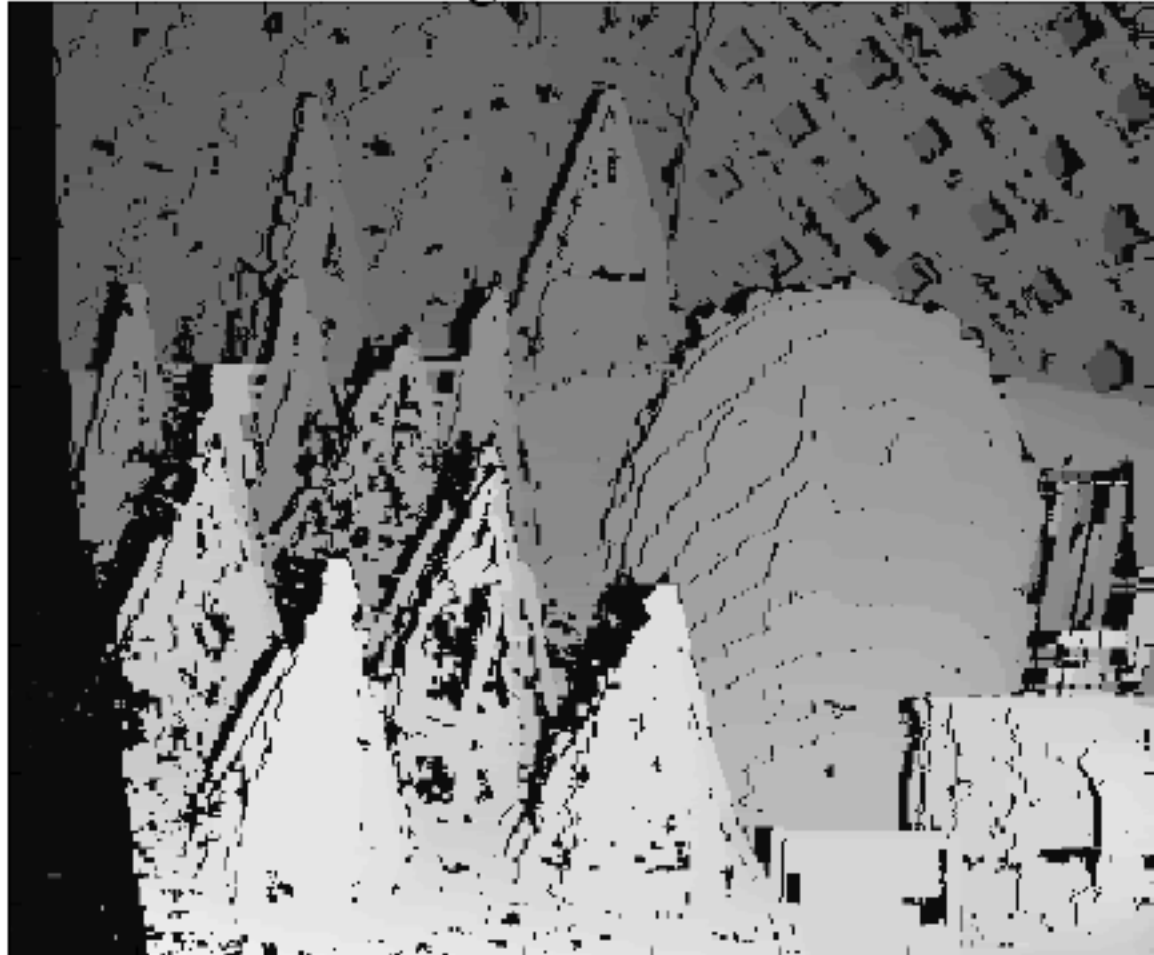
DSI and Scanline Consistency

Assigning disparities to all pixels in left scanline now
amounts to finding a connected path through the DSI



Example

Result of DP alg



Result without DP (independent pixels)



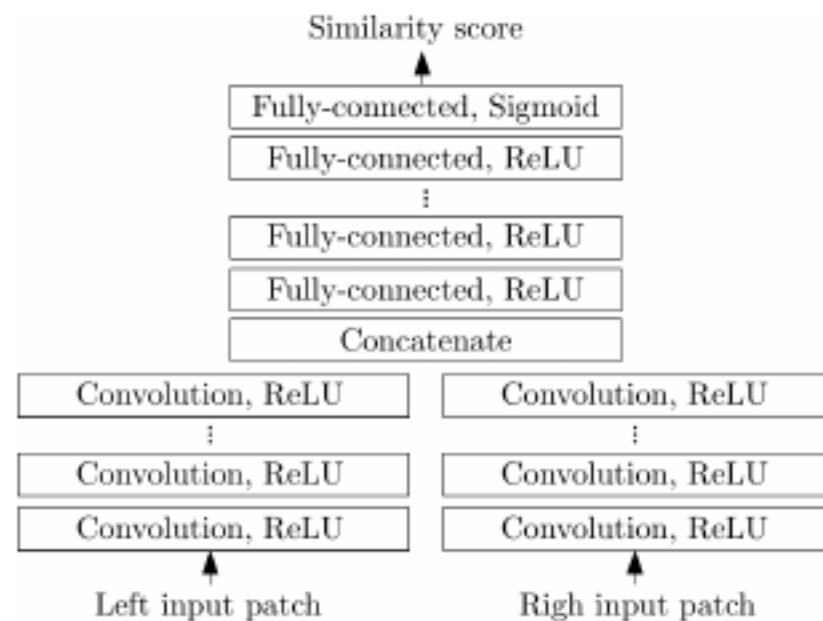
Result of DP alg. Black pixels = occluded.

**Last week: Siamese networks
for stereo matching**

Siamese Network Architecture

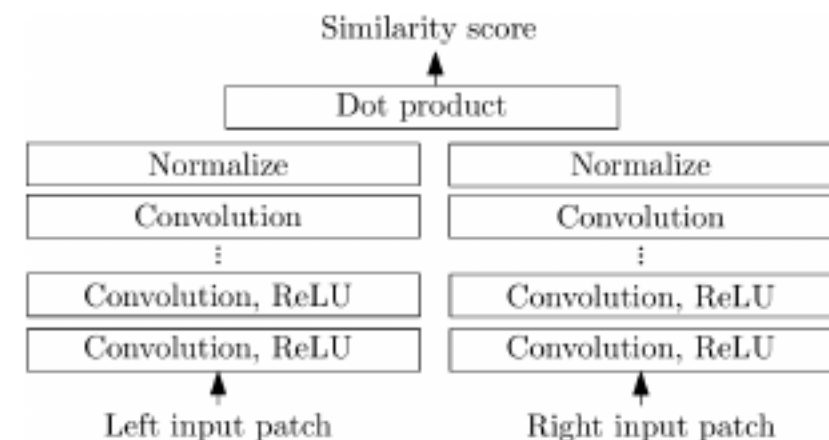
Learned Similarity:

- ▶ Learn features & sim. metric
- ▶ Potentially more expressive
- ▶ Slow (WxHxD MLP evaluations)

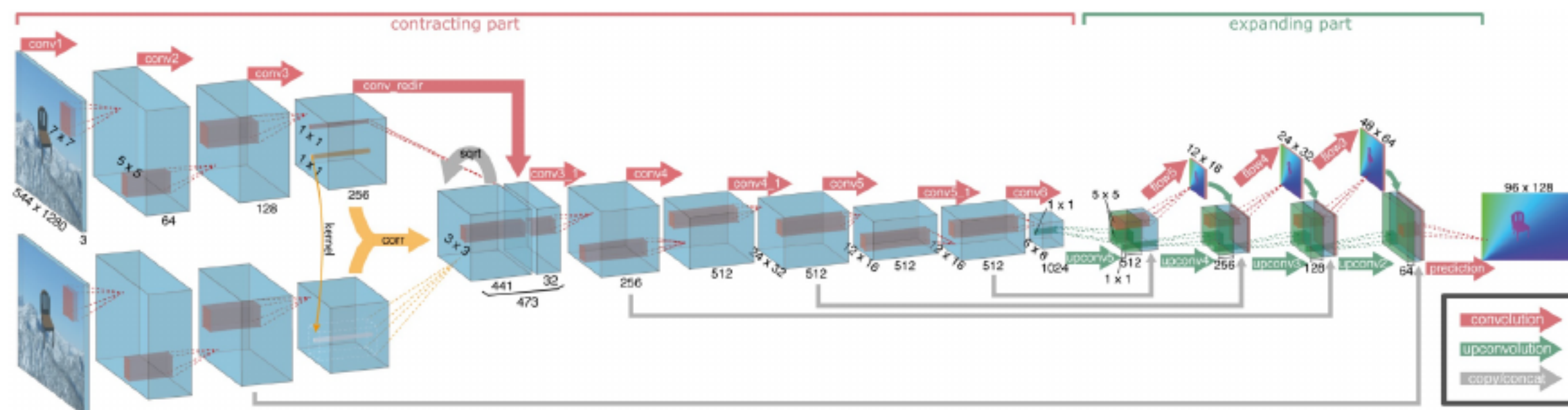


Cosine Similarity:

- ▶ Learn features & apply dot-product
- ▶ Features must do the heavy lifting
- ▶ Fast matching (no network eval.)



DispNet



- ▶ DispNet was the first end-to-end trained deep neural network for stereo
- ▶ It used a U-Net like architecture with skip-Connections to retain details
- ▶ Correlation layer (40px displacement corresponding to 160px in input image)
- ▶ Multi-scale loss (disparity error in pixels), curriculum learning (learn easy-to-hard)

Mayer et al.: A Large Dataset to Train Convolutional Networks for Disparity, Optical Flow, and Scene Flow Estimation. CVPR, 2016.

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Graphical models for structured prediction

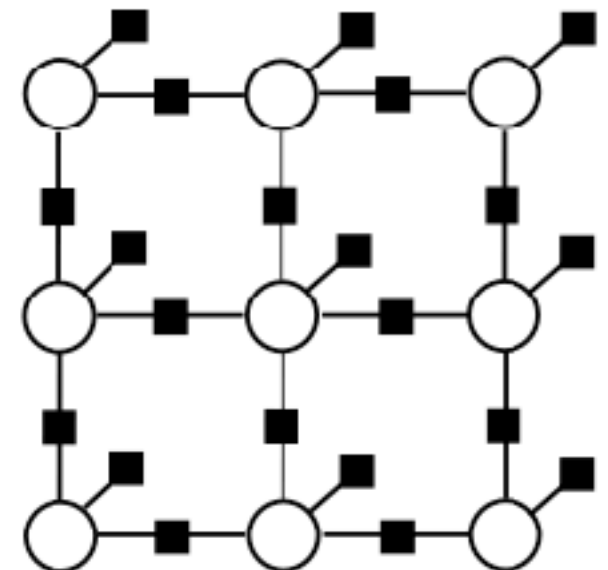
Basic idea

- We want to find a depth at each pixel by matching it to the corresponding point in the other image
- The match should optimize similarity
 - this is an independent objective at each pixel
- At the same time, matches of neighboring pixels should usually lead to similar depth estimates
 - this objective refers to results of pairs of neighboring pixels

A graphical model (of order 2)

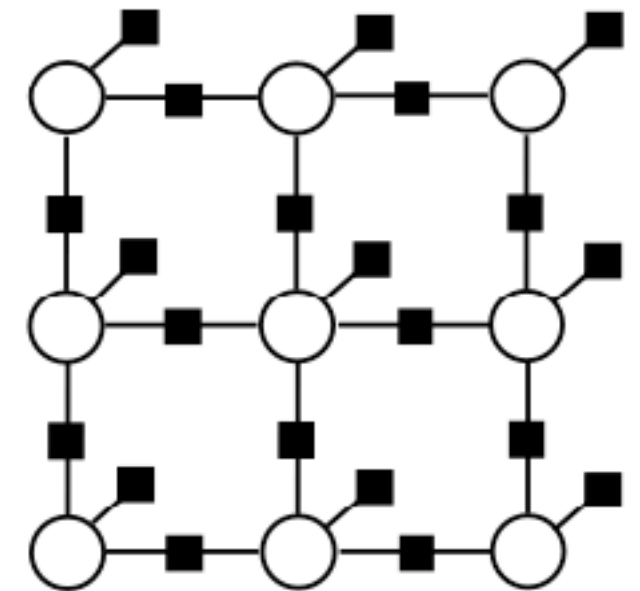
- Dependencies of parameters are modeled as a graph
 - here corresponding to 4-neighborhoods on the pixel grid
- We find a solution minimizing both
 - the **data cost** at each node (=pixel)
 - the **smoothness cost** at each edge (=neighboring pixel pair)
- This is modeled as an exponential probability distribution over unary and binary terms connected to the graph

$$p(\mathbf{D}) \propto \exp \left\{ - \sum_i \psi_{data}(d_i) - \lambda \sum_{i \sim j} \psi_{smooth}(d_i, d_j) \right\}$$



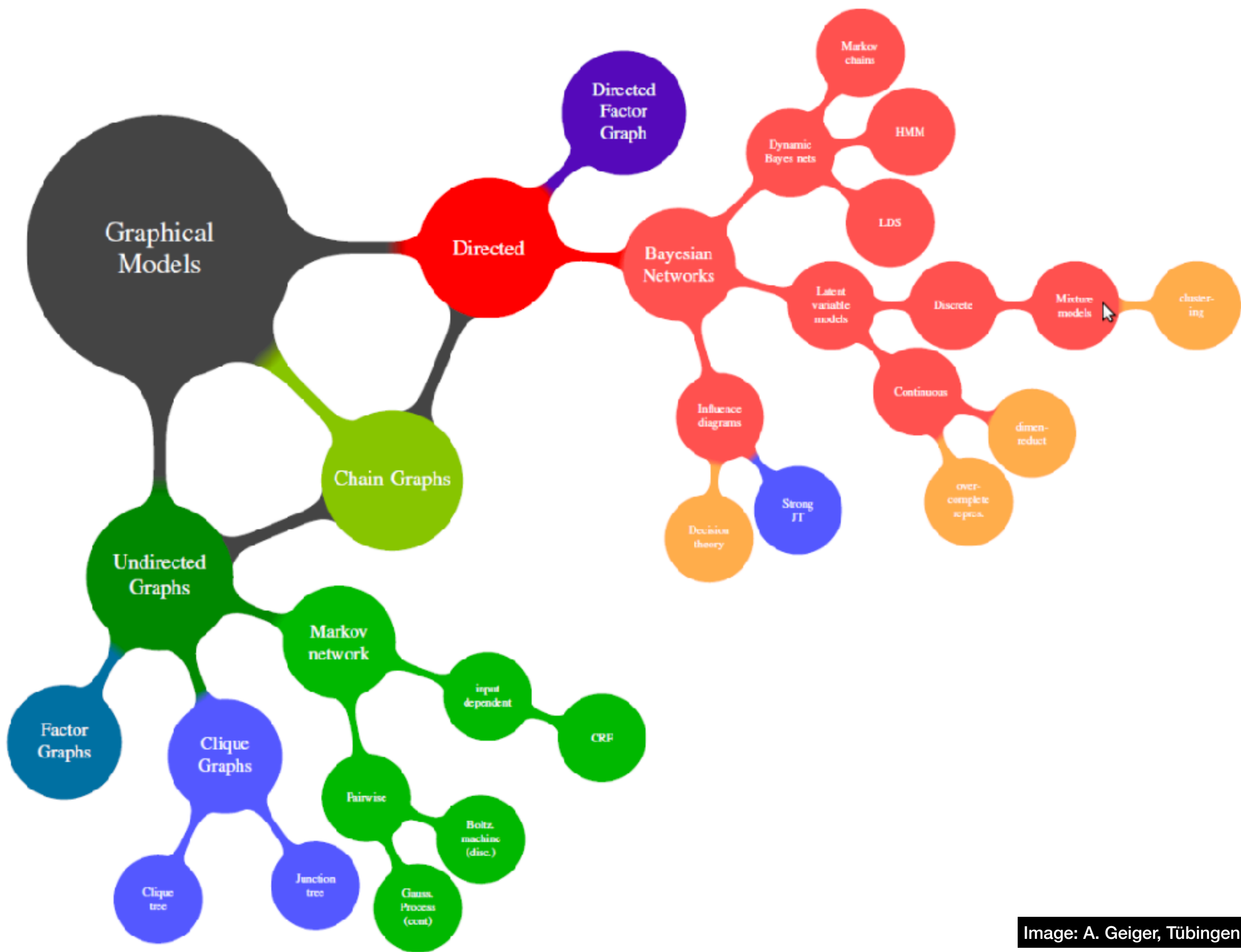
A graphical model (of order 2)

$$p(\mathbf{D}) \propto \exp \left\{ - \sum_i \psi_{data}(d_i) - \lambda \sum_{i \sim j} \psi_{smooth}(d_i, d_j) \right\}$$



A graphical model (of order 2)

- View the solution as a maximum a posteriori estimate of a conditional probability distribution
- The dependency structure is modelled as a graph, based on local constraints
- The output is a complex object („structured“), ie. an image (as opposed to a class label or regression value)
- Optimal solutions exist for certain problem classes, often approximate solutions
- „Interpretable by design“!

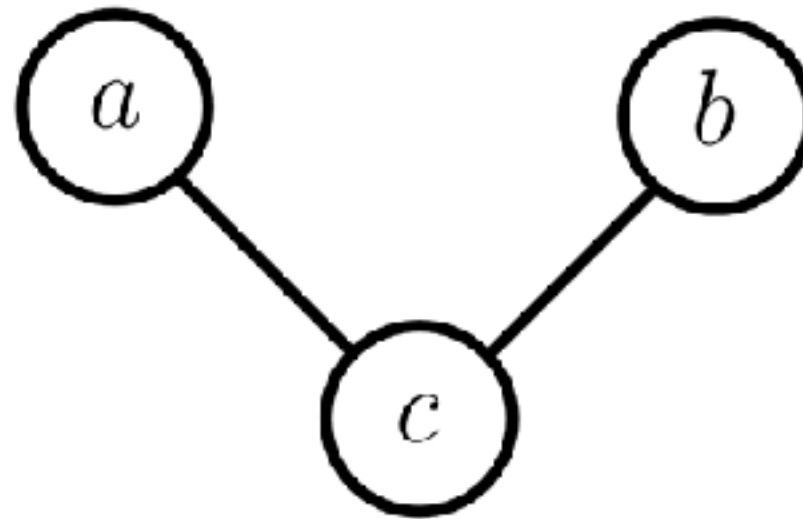


Markov Random Fields

- Potentials $\phi(x)$ are non-negative functions of a variable x
- $\phi(x, y)$ is the joint potential of two variables
- Given a set of variables $X = \{x_1, \dots, x_N\}$ and a set of maximal cliques $\{X_k\}_{k=1}^K$ over the graph, the Markov Random Field is the product over potentials

$$p(X) = \frac{1}{Z} \prod_{k=1}^K \phi_k(X_k)$$

Properties of Markov Random Fields



$$p(a, b, c) = \frac{1}{Z} \phi_1(a, c) \phi_2(b, c)$$

- ▶ Two maximal cliques of size two: $\mathcal{X}_1 = \{a, c\}$ and $\mathcal{X}_2 = \{b, c\}$
- ▶ Z normalizes the distribution and is called **partition function**

$$Z = \sum_{a,b,c} \phi_1(a, c) \phi_2(b, c)$$

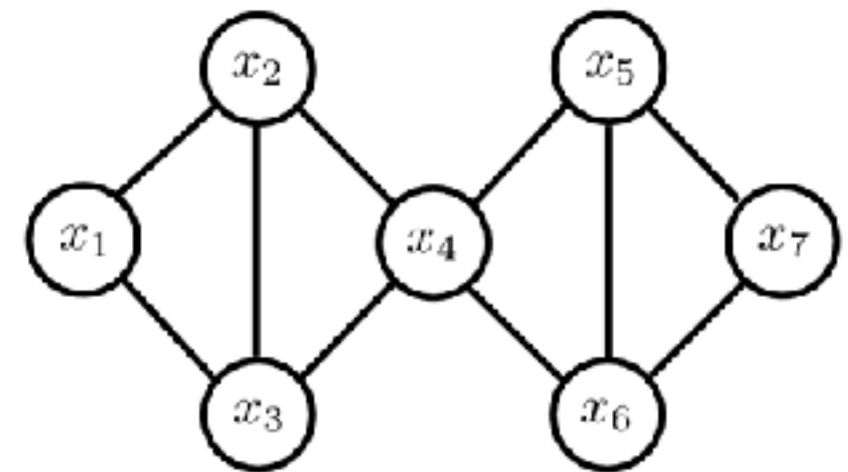
Global Markov property

Separation

A subset $\mathcal{S}$ separates $\mathcal{A}$ from $\mathcal{B}$ if every path from a member of $\mathcal{A}$ to any member of $\mathcal{B}$ passes through $\mathcal{S}$.

Global Markov Property

For disjoint sets of variables $(\mathcal{A}, \mathcal{B}, \mathcal{S})$ where $\mathcal{S}$ separates $\mathcal{A}$ from $\mathcal{B}$, we have $\mathcal{A} \perp\!\!\!\perp \mathcal{B} \mid \mathcal{S}$



Local Markov property

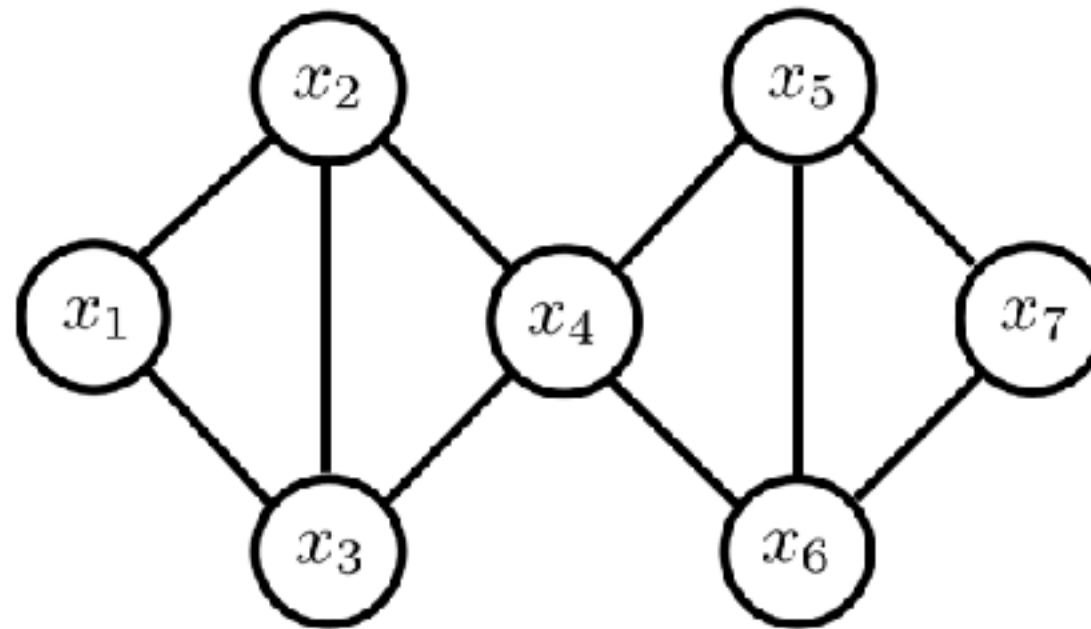
Local Markov Property

When conditioned on its neighbors, x becomes independent of the remaining variables of the graph:

$$p(x \mid \mathcal{X} \setminus \{x\}) = p(x \mid ne(x))$$

- ▶ The set of neighboring nodes $ne(x)$ is called **Markov blanket**
- ▶ This also holds for sets of variables

Local Markov property



- ▶ $p(x_4 \mid x_1, x_2, x_3, x_5, x_6, x_7) = p(x_4 \mid x_2, x_3, x_5, x_6)$
- ▶ In other words $x_4 \perp\!\!\!\perp \{x_1, x_7\} \mid \{x_2, x_3, x_5, x_6\}$
- ▶ Similarly, other independence relationships can be read off the graph

Hammersley-Clifford Theorem

Hammersley-Clifford Theorem

A probability distribution that has a **strictly positive mass** or density satisfies the **Markov properties** with respect to an undirected graph $\mathcal{G}$ if and only if it is a Gibbs random field, i.e., its density can be **factorized** over the (maximal) cliques of the graph.

Factorization induced by a graph is not unique

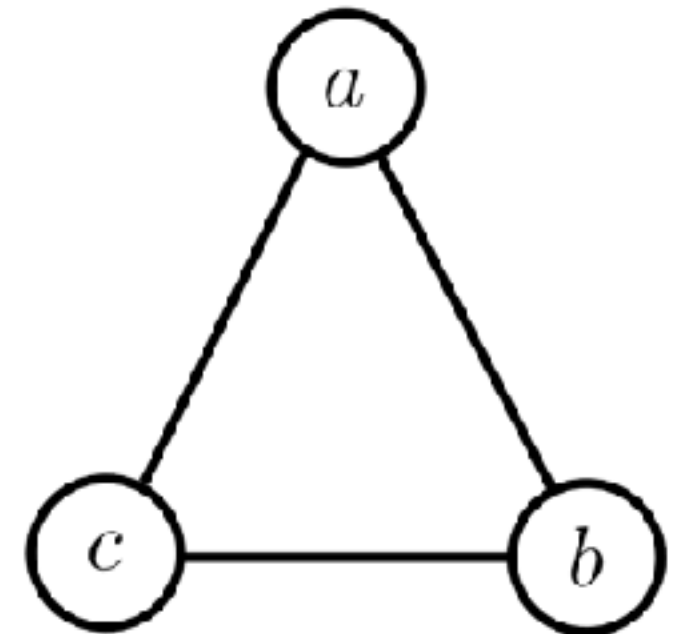
- Consider this factorization into potential functions:

$$p(a, b, c) = \frac{1}{Z} \phi(a, b) \phi(b, c) \phi(c, a)$$

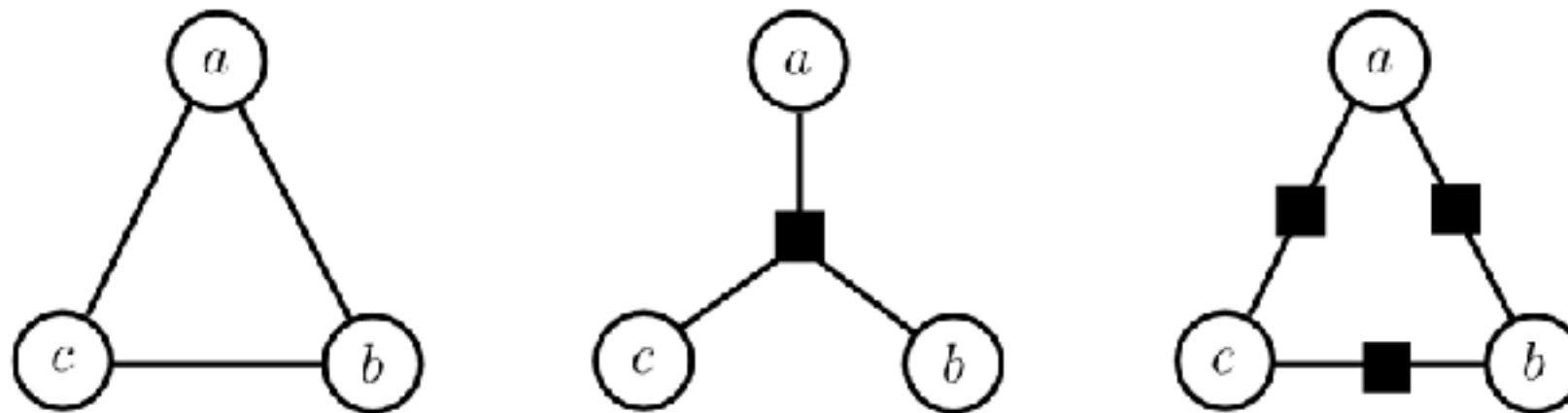
- Which other factorization is represented by this Markov network?

$$p(a, b, c) = \frac{1}{Z} \phi(a, b, c)$$

- What is the corresponding Markov network / MRF?



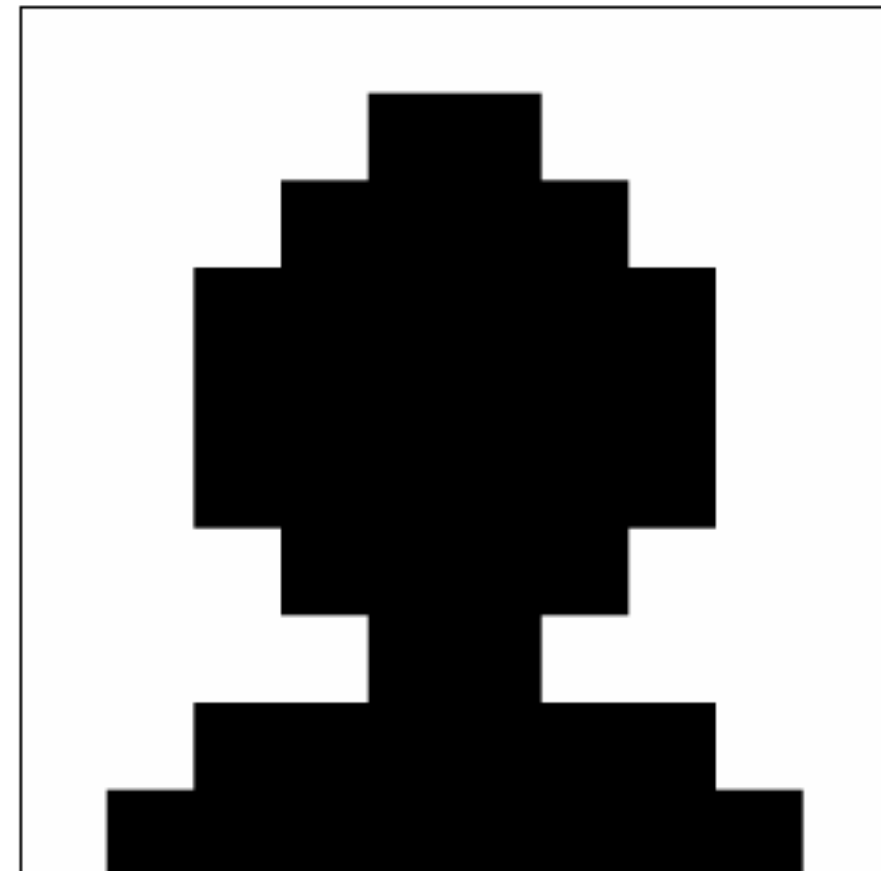
Factor graphs



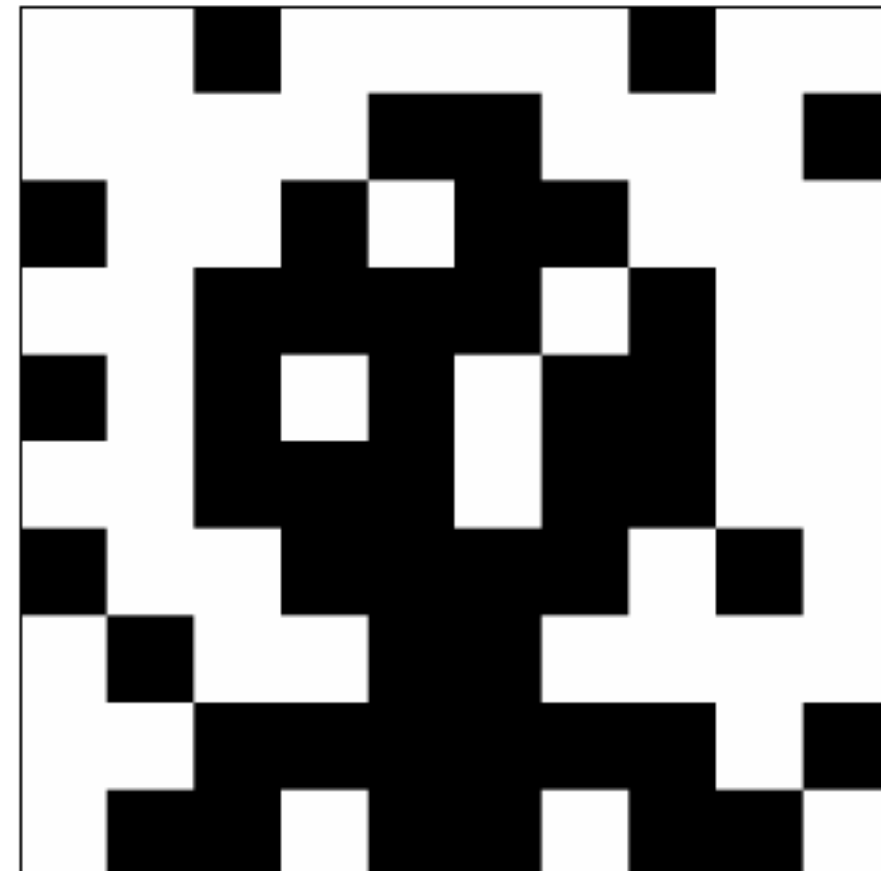
- ▶ Left: Markov Network of $\frac{1}{Z}\phi(a, b, c)$ and $\frac{1}{Z}\phi(a, b)\phi(b, c)\phi(c, a)$
- ▶ Middle: Factor graph representation of $\frac{1}{Z}\phi(a, b, c)$
- ▶ Right: Factor graph representation of $\frac{1}{Z}\phi(a, b)\phi(b, c)\phi(c, a)$
- ▶ The two factor graphs correspond to the **same Markov network**
- ▶ But they allow to distinguish different **factorizations** by making them **explicit**

Example: Image restoration

Example: Image restoration

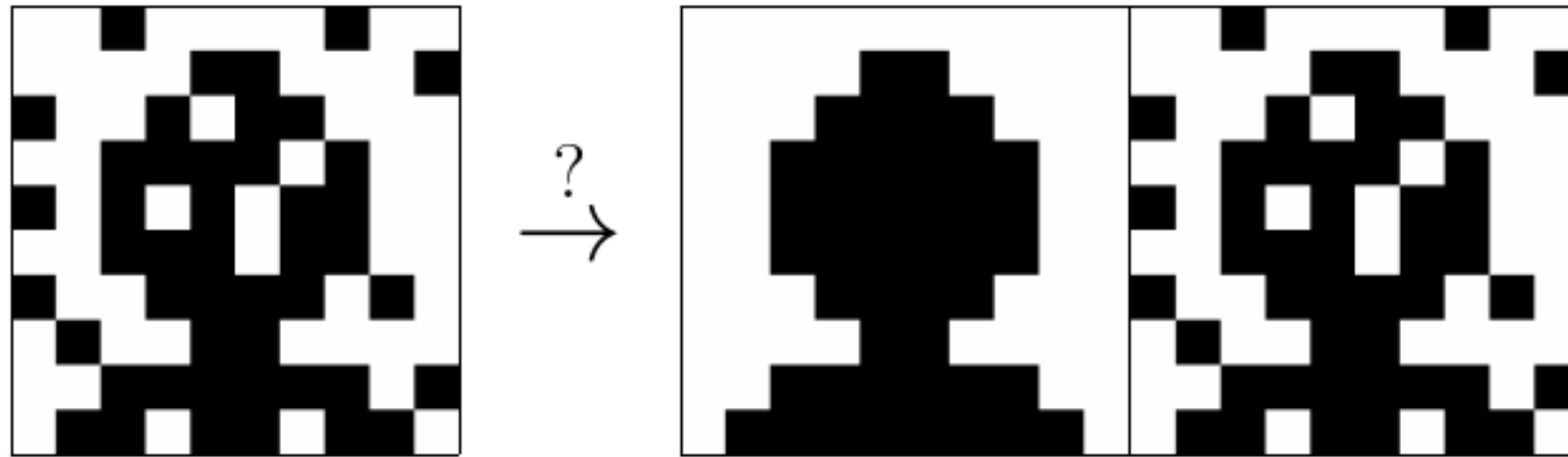


Example: Image restoration

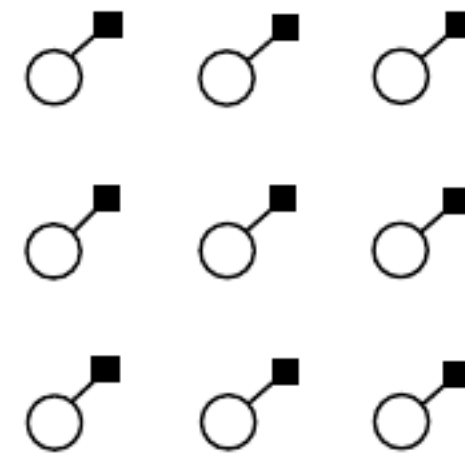


Example: Image restoration

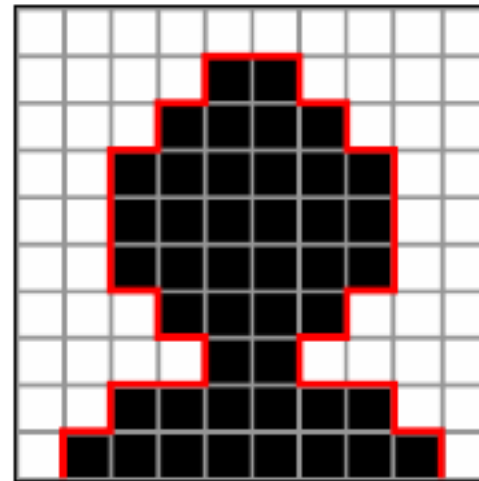
Can we recover the original image from a noisy observation?



- ▶ Variables: $x_1, \dots, x_{100} \in \{0, 1\}$
- ▶ Unary log factors: $\psi_1(x_1), \dots, \psi_{100}(x_{100})$
- ▶ $\psi_i(x_i) = [x_i = o_i]$ with observation o_i
- ▶ Log representation: $\psi_i(x_i) = \log f_i(x_i)$
 $p(x) = \frac{1}{Z} \prod_i f_i(x_i) = \frac{1}{Z} \exp \{ \sum_i \psi_i(x_i) \}$

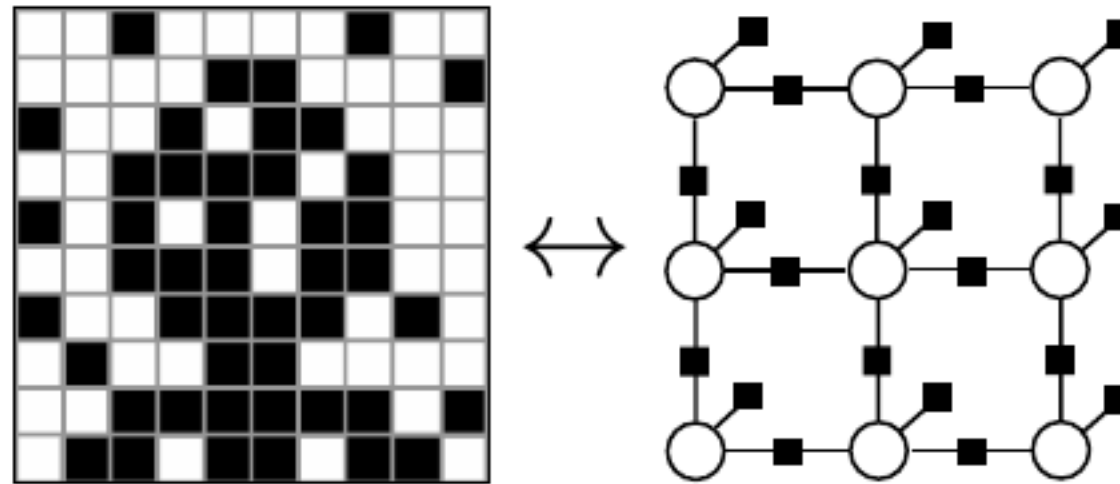


Example: Image restoration



- ▶ We like to integrate prior knowledge by adding constraints to the problem
- ▶ What prior knowledge do we have about this image?
- ▶ Smoothness: Neighboring pixels tend to have the same label
- ▶ How many neighbors share the same label?
- ▶ $10 \times 10 \times 2 - 20 = 180$ neighborhood relationships in total
- ▶ 34x label transition and 146x same label (factor 4.3 more)

Example: Image restoration



$$p(x_1, \dots, x_{100}) = \frac{1}{Z} \exp \left\{ \sum_{i=1}^{100} \psi_i(x_i) + \sum_{i \sim j} \psi_{ij}(x_i, x_j) \right\}$$

- Unary log factors: $\psi_i(x_i) = [x_i = o_i]$ with observation $o_i \in \{0, 1\}$
- Pairwise log factors: $\psi_{ij}(x_i, x_j) = \lambda \cdot [x_i = x_j]$
- Parameter λ controls strength of prior $\Rightarrow$ Exercise; Next time: Stereo

Back to stereo matching

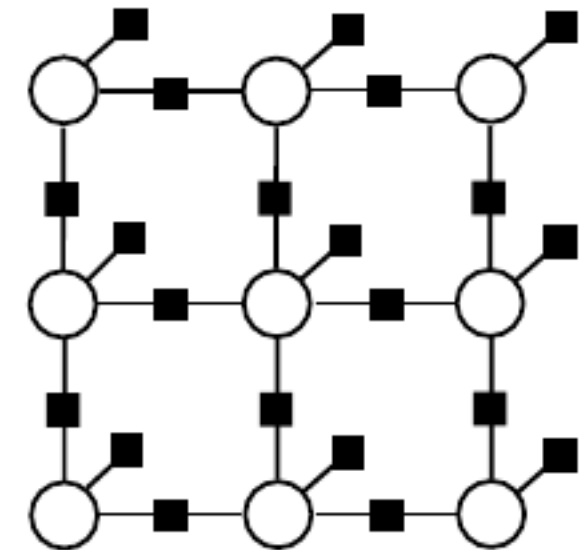
Dense Stereo MRF

Specify a loopy **Markov Random Field (MRF)** on a grid and solve for the whole disparity map $\mathbf{D}$ at once (MAP solution = minimum of energy):

$$p(\mathbf{D}) \propto \prod_i f_{data}(d_i) \times \prod_{i \sim j} f_{smooth}(d_i, d_j)$$

Or equivalently:

$$p(\mathbf{D}) \propto \exp \left\{ - \underbrace{\left(\sum_i \psi_{data}(d_i) + \lambda \sum_{i \sim j} \psi_{smooth}(d_i, d_j) \right)}_{\text{Energy } E} \right\}$$

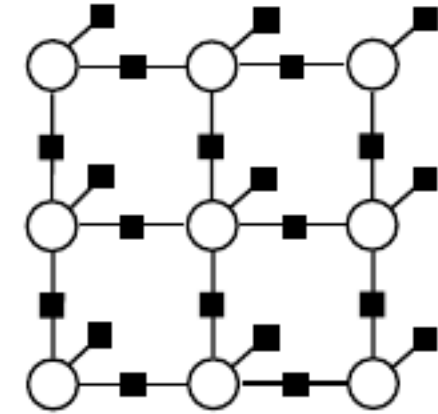


- ▶ $\psi(\cdot) := -\log f(\cdot)$, $i \sim j$: neighboring pixels on a 4-connected grid
- ▶ $\psi_{data}(d)$: unary terms, and $\psi_{smooth}(d, d')$ pairwise terms

Dense Stereo MRF

Markov Random Field (MRF) from last slide:

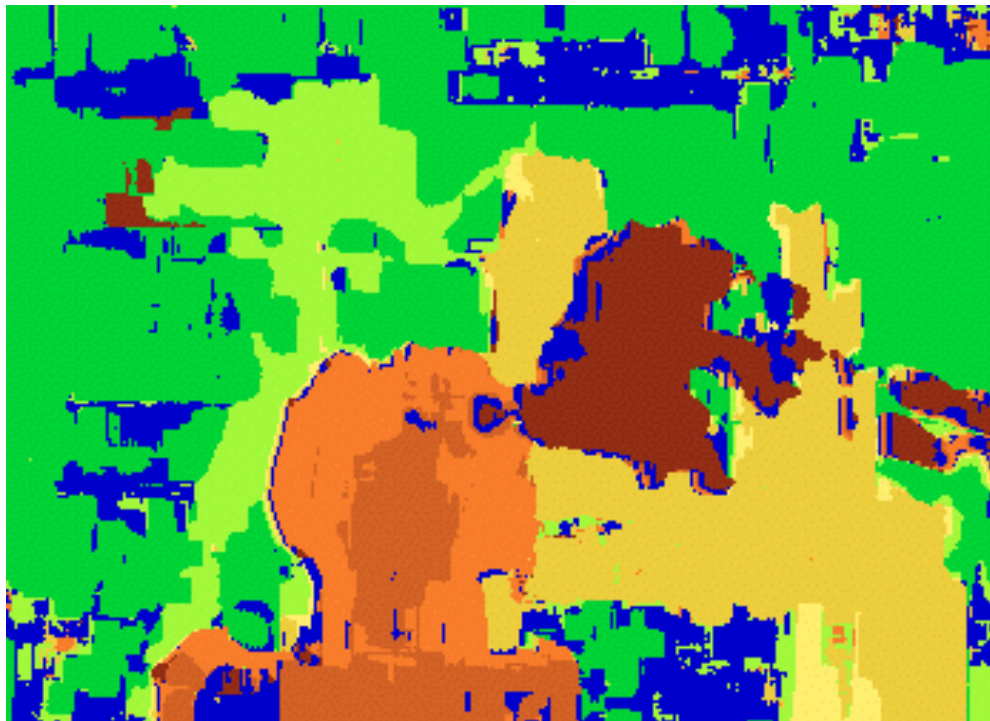
$$p(\mathbf{D}) \propto \exp \left\{ - \underbrace{\left(\sum_i \psi_{data}(d_i) + \lambda \sum_{i \sim j} \psi_{smooth}(d_i, d_j) \right)}_{\text{Energy } E} \right\}$$



- ▶ $i \sim j$: neighboring pixels on a 4-connected grid
- ▶ **Unary terms:** Matching cost $\psi_{data}(d)$
- ▶ **Pairwise terms:** Encourages smoothness between adjacent pixels, e.g.:
 - ▶ Potts model: $\psi_{smooth}(d, d') = [d \neq d']$
 - ▶ Truncated l_1 penalty: $\psi_{smooth}(d, d') = \min(|d - d'|, \tau)$
- ▶ The parameter λ is added to control the strength of the smoothness prior
- ▶ Solve MRF approximately for disparity map $\mathbf{D}$ using BP, graph cuts, etc.



Block matching result



Window-based matching
(best window size)



Ground truth

MRF result (via graph cuts)



Graph cuts-based method



Ground truth

Boykov et al., [Fast Approximate Energy Minimization via Graph Cuts](#),

International Conference on Computer Vision 1999.

Summary

- ▶ Block matching suffers from **ambiguities**
- ▶ Choosing window size is problematic (tradeoff)
- ▶ Incorporating **smoothness constraints** can resolve some of the ambiguities and allows for choosing small windows (no bleeding artifacts)
- ▶ Can be formulated as **MAP inference in a discrete MRF**
- ▶ MAP solution can be obtained using belief propagation, graph cuts, etc.
- ▶ Integrating recognition cues can further regularize the problem